System Domain

1. System Context

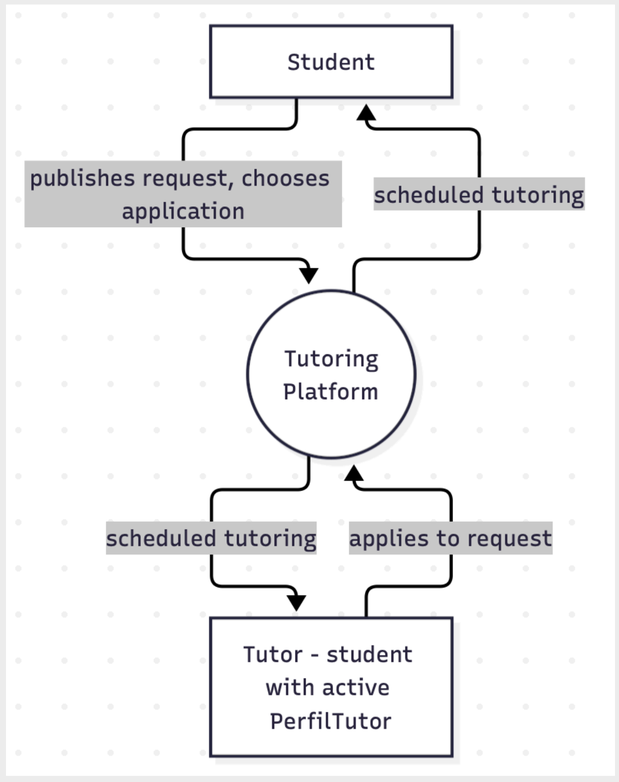
The system is a web and mobile platform where students of the institution find academic support from other students. Each student has a single institutional account; from it they can request tutoring and, optionally, activate a tutor profile (PerfilTutor) without creating another account. The flow is request-based: a student publishes a tutoring request, tutors apply to it by accepting its conditions, the student chooses one application, and the tutoring session is scheduled.

Each request states the subject, topic, date, start and end time, modality, place (if in person) and the TOTAL budget in COP. When a session is confirmed, the system checks that neither participant has another tutoring session at that time.

Problem: students struggle to find tutors who fit their subject, schedule and budget, and to coordinate sessions in an organized way.

In scope: institutional accounts, tutor profile activation, publishing requests, tutor applications, choosing an application, scheduling tutoring sessions and the schedule-conflict check.

Out of scope: reviews, chat, notifications and integrated payments.



2. Actor Characterization

Actor	Description	Main responsibilities
Student	Holder of an institutional account who needs academic support. Is the default role of every user.	Publish tutoring requests with all required data, review the applications received, choose one, and attend
Tutor	A student who has activated a PerfilTutor on the same account. Can still act as a student.	Apply to requests by accepting their conditions, and attend the scheduled session.

3. Business Rules

ID	Business rule
BR-01	Each student has exactly one institutional account; becoming a tutor never requires a second account.
BR-02	A student can activate a PerfilTutor on their account and then act both as requester and as tutor.
BR-03	A request must indicate subject, topic, date, start time, end time, modality, and budget.
BR-04	If the modality is in person, the request must also indicate the place.
BR-05	The budget is a single TOTAL amount in COP for the whole session, not a rate per hour.
BR-06	Only users with an active PerfilTutor can apply to a request.
BR-07	A tutor applies to a request by accepting its conditions (schedule, modality, place and budget).
BR-08	A user cannot apply to their own request.
BR-09	The student chooses one of the applications received, and the tutoring session is then scheduled.
BR-10	When the session is confirmed, the system must verify that neither the student nor the tutor has another tutoring session in that time slot; if there is a conflict, the confirmation is not allowed.

Tarjetas CRC
Núcleo del flujo de tutorías

Clase	Responsabilidades	Colaboradores
Estudiante	Mantener identidad institucional, intereses y rol activo. Asociar un perfil de tutor opcional. Gestionar solicitudes propias y consultar sus tutorías.	PerfilTutor SolicitudTutoria Tutoria
PerfilTutor	Mantener materias, presentación y disponibilidad. Crear o retirar postulaciones compatibles. Consultar las tutorías que imparte.	Estudiante Postulacion Tutoria
SolicitudTutoria	Conservar materia, tema, fecha, horario, modalidad, lugar y presupuesto total. Recibir postulaciones mientras esté abierta. Permitir al autor aceptar una; cerrar las demás. Gestionar cancelación y vencimiento.	Estudiante Postulacion Tutoria
Postulacion	Vincular un tutor con una solicitud abierta. Confirmar el horario y precio publicados. Evitar duplicados y postulaciones a solicitudes propias. Mantener su estado: pendiente, aceptada, rechazada o retirada.	PerfilTutor SolicitudTutoria
Tutoria	Crearse a partir de la postulación aceptada. Fijar participantes, horario, precio y lugar o enlace. Evitar cruces de agenda de ambos participantes. Registrar finalización o cancelación y liberar el horario.	Estudiante PerfilTutor SolicitudTutoria Postulacion

Class Diagram

Estudiante

- id: int
- nombre: String
- correoInstitucional: String
- claveHash: String
- correoVerificado: Boolean
- rolActivo: Rol
- intereses: List<String>

+ actualizarPerfil(nombre: String, intereses: List<String>): void
+ activarPerfilTutor(): PerfilTutor
+ cambiarRol(rol: Rol): void
+ publicarSolicitud(s: SolicitudTutoria): void
+ consultarTutorias(): List<Tutoria>
+ estaDisponible(inicio: DateTime, fin: DateTime): Boolean

PerfilTutor

- id: int
- presentacion: String
- materias: List<String>
- disponibilidad: List<FranjaHoraria>
- activo: Boolean

+ actualizarPresentacion(texto: String): void
+ definirMaterias(materias: List<String>): void
+ definirDisponibilidad(franjas: List<FranjaHoraria>): void
+ postularse(s: SolicitudTutoria): Postulacion
+ consultarTutorias(): List<Tutoria>

Estudiante 1 — habilita — 0..1 PerfilTutor

Estudiante 1 — publica — 0..* SolicitudTutoria

PerfilTutor 1 — realiza — 0..* Postulacion

Estudiante 1 — recibe como aprendiz — 0..* Tutoria

PerfilTutor 1 — imparte — 0..* Tutoria

SolicitudTutoria

- id: int
- materia: String
- tema: String
- inicio: DateTime
- fin: DateTime
- modalidad: Modalidad
- lugar: String [0..1]
- presupuestoTotal: Decimal
- publicadaEn: DateTime
- estado: EstadoSolicitud

+ validar(): Boolean
+ recibirPostulacion(p: Postulacion): void
+ aceptar(p: Postulacion): Tutoria
+ cancelar(): void
+ vencer(): void

Postulacion

- id: int
- creadaEn: DateTime
- estado: EstadoPostulacion

+ validar(): Boolean
+ retirar(): void
+ marcarAceptada(): void
+ rechazar(): void

SolicitudTutoria 1 — recibe — 0..* Postulacion

Postulacion 1 — origina — 0..1 Tutoria

Tutoria

- id: int
- inicio: DateTime
- fin: DateTime
- precioTotal: Decimal
- modalidad: Modalidad
- lugarOEnlace: String [0..1]
- estado: EstadoTutoria

+ definirEncuentro(lugarOEnlace: String): void
+ seCruza(inicio: DateTime, fin: DateTime): Boolean
+ finalizar(): void
+ cancelar(): void

TIPOS DE APOYO

Rol: APRENDIZ | TUTOR
Modalidad: PRESENCIAL | VIRTUAL
EstadoSolicitud: ABIERTA | ASIGNADA | CANCELADA | VENCIDA
EstadoPostulacion: PENDIENTE | ACEPTADA | RECHAZADA | RETIRADA
EstadoTutoria: PROGRAMADA | FINALIZADA | CANCELADA
FranjaHoraria: {inicio: DateTime, fin: DateTime}
Decimal: dinero en COP; no usar float.

REGLAS AL CONFIRMAR

Solo el autor elige un tutor distinto de sí mismo.
Solicitud abierta y postulación pendiente de ella.
Una postulación por tutor y solicitud.
Se acepta una y se rechazan las demás pendientes.
Se crea la tutoría con las condiciones publicadas.
Se comprueba la agenda de ambas personas.
Todo se guarda en una misma transacción.

− atributo privado + método público 1 uno 0..1 opcional 0..* cero o muchos ◆ composición

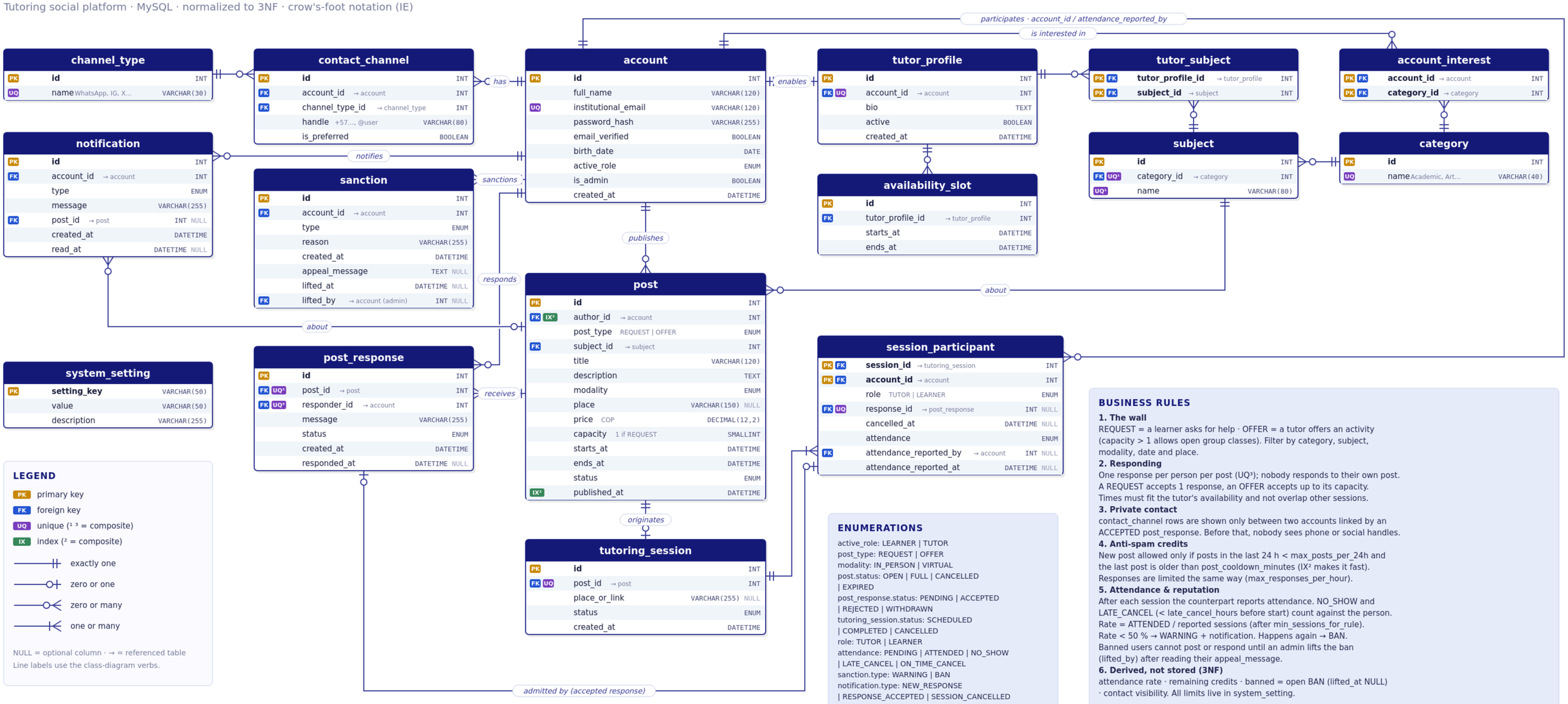
Las líneas representan referencias entre objetos. No se duplican aquí como atributos estudianteId, tutorId, etc.

Modelo inicial: 5 clases; una cuenta con dos roles, condiciones fijas y sin pagos integrados, chat o reseñas.

Relational Database Model

Relational Database Model

Tutoring social platform · MySQL · normalized to 3NF · crow's-foot notation (IE)



LEGEND

- PK primary key
- FK foreign key
- UQ unique (¹ ³ = composite)
- IX index (² = composite)
- exactly one
- zero or one
- zero or many
- one or many

NULL = optional column · → = referenced table
Line labels use the class-diagram verbs.

BUSINESS RULES

1. The wall
REQUEST = a learner asks for help · OFFER = a tutor offers an activity (capacity > 1 allows open group classes). Filter by category, subject, modality, date and place.
2. Responding
One response per person per post (UQ³); nobody responds to their own post. A REQUEST accepts 1 response, an OFFER accepts up to its capacity. Times must fit the tutor's availability and not overlap other sessions.
3. Private contact
contact_channel rows are shown only between two accounts linked by an ACCEPTED post_response. Before that, nobody sees phone or social handles.
4. Anti-spam credits
New post allowed only if posts in the last 24 h < max_posts_per_24h and the last post is older than post_cooldown_minutes (IX² makes it fast). Responses are limited the same way (max_responses_per_hour).
5. Attendance & reputation
After each session the counterpart reports attendance. NO_SHOW and LATE_CANCEL (< late_cancel_hours before start) count against the person. Rate = ATTENDED / reported sessions (after min_sessions_for_rule). Rate < 50 % → WARNING + notification. Happens again → BAN. Banned users cannot post or respond until an admin lifts the ban (lifted_by) after reading their appeal_message.
6. Derived, not stored (3NF)
attendance rate · remaining credits · banned = open BAN (lifted_at NULL) · contact visibility. All limits live in system_setting.

ENUMERATIONS

active_role: LEARNER | TUTOR
post_type: REQUEST | OFFER
modality: IN_PERSON | VIRTUAL
post.status: OPEN | FULL | CANCELLED | EXPIRED
post_response.status: PENDING | ACCEPTED | REJECTED | WITHDRAWN
tutoring_session.status: SCHEDULED | COMPLETED | CANCELLED
role: TUTOR | LEARNER
attendance: PENDING | ATTENDED | NO_SHOW | LATE_CANCEL | ON_TIME_CANCEL
sanction.type: WARNING | BAN
notification.type: NEW_RESPONSE | RESPONSE_ACCEPTED | SESSION_CANCELLED | ATTENDANCE_WARNING | BANNED
Money: DECIMAL(12,2) in COP, never FLOAT.

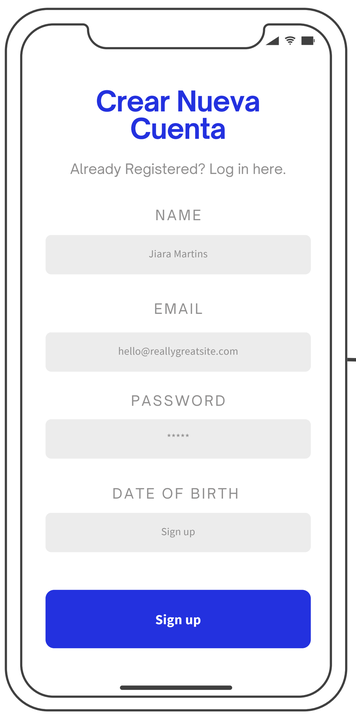
CLASS DIAGRAM → TABLES

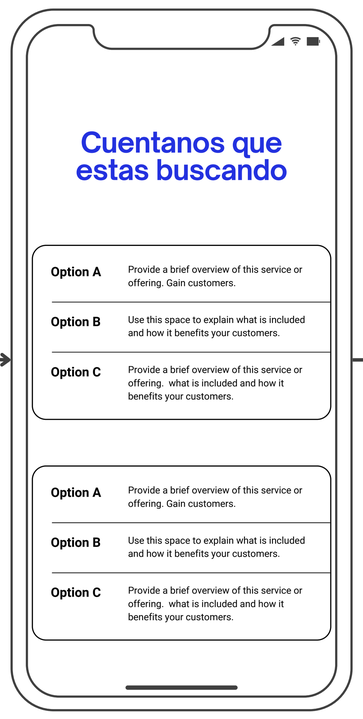
Estudiante → account (intereses → account_interest)
PerfilTutor → tutor_profile (materias → tutor_subject, disponibilidad / FranjaHoraria → availability_slot)
SolicitudTutoria → post with post_type = REQUEST (OFFER is new: tutors publish activities)
Postulacion → post_response (also used by learners to join an OFFER)
Tutoria → tutoring_session + session_participant (group classes and per-person attendance)
New for the MVP: contact_channel, channel_type, category, subject, sanction, notification, system_setting

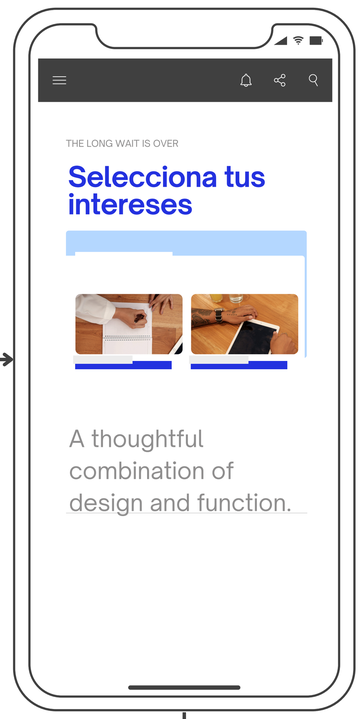
Wireframe Human inteligence

Mobile

Let's make some decisions about our application's interface.



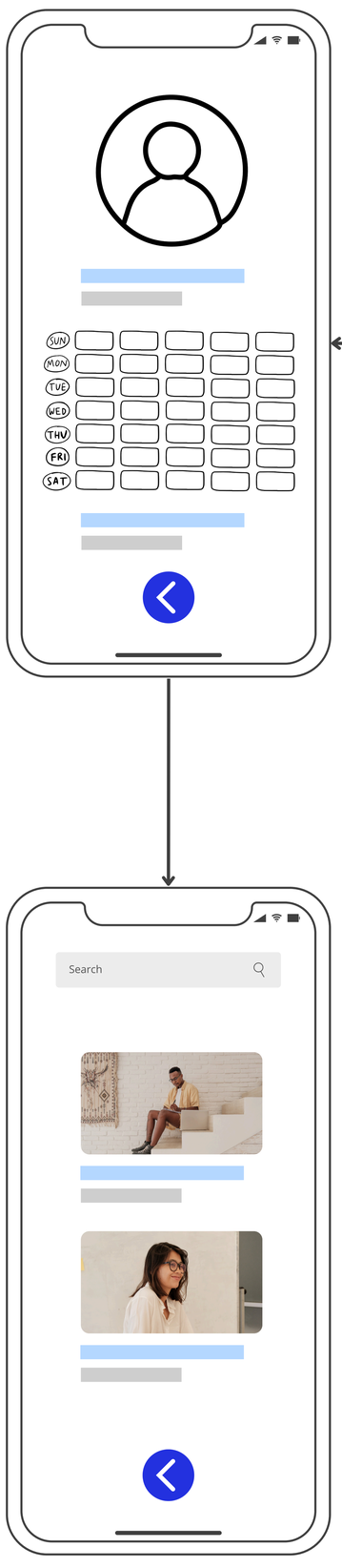




Vista como Mentor

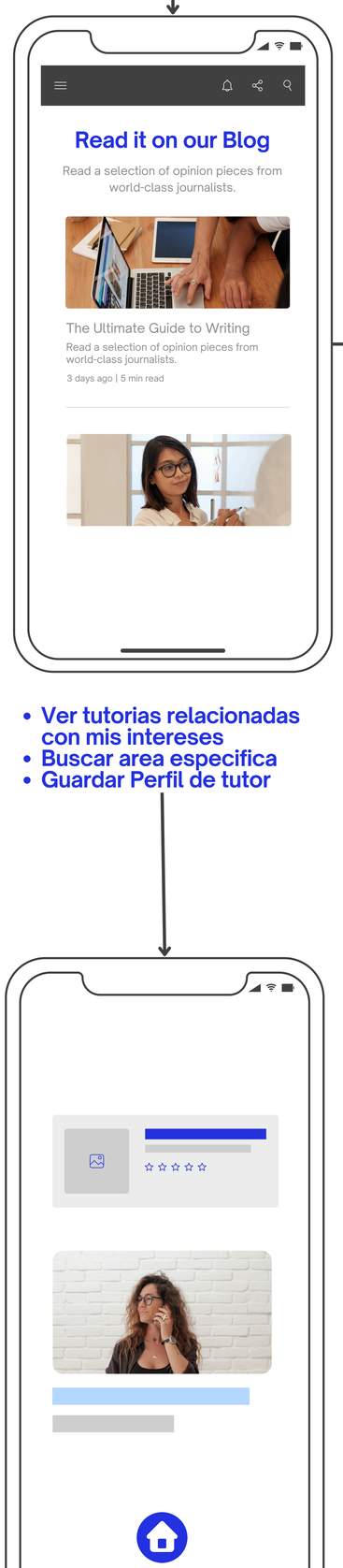
Vista como Aprendiz

- Mi Cuenta
- Calendario
- Tutorias Programadas
- Solicitudes Pendientes
- Mis Reseñas
- Ganancias

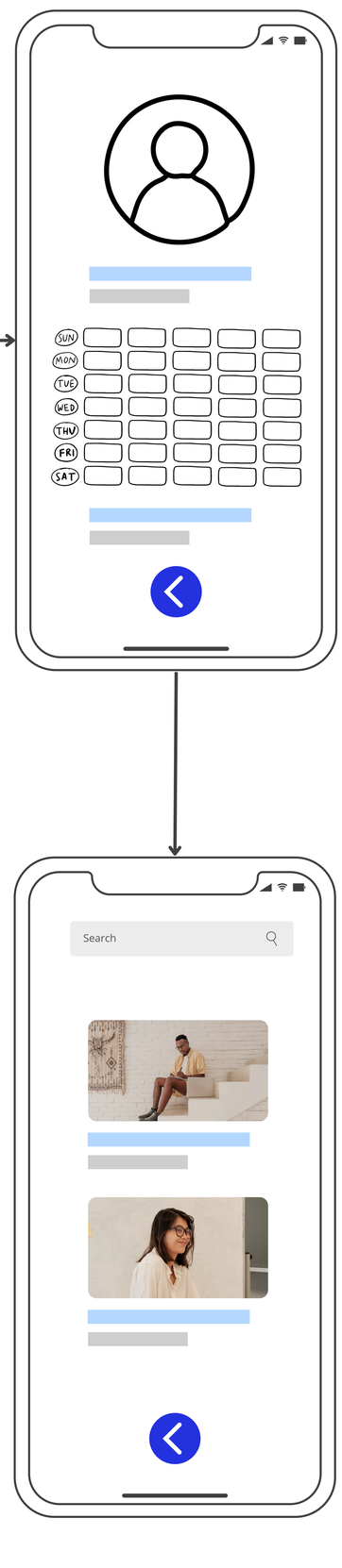




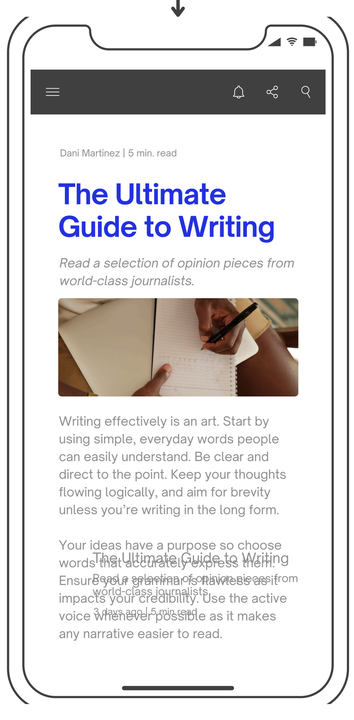
- Crear Nueva Tutoria
- Añadir Titulo
- Añadir descripción
- Añadir foto
- Añadir Ubicación
- Anadir costo de la tutoria
- Guardar y Publicar
- Datos de contacto

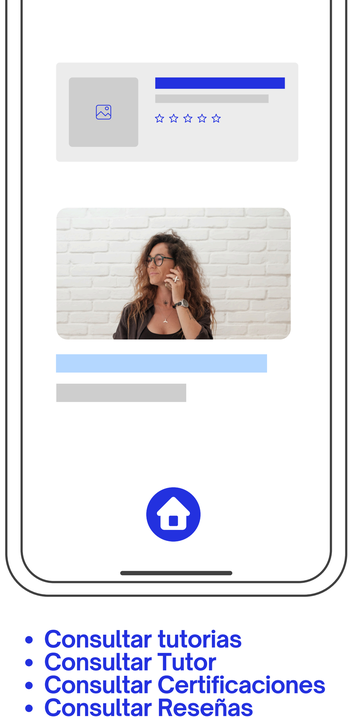


- Mi Cuenta
- Calendario
- Tutorias Programadas
- Mis Reseñas



- Ver tutorias relacionadas con mis intereses
- Buscar area especifica
- Guardar Perfil de tutor





- Consultar tutorias confirmadas
- Consultar tutorias Pendientes
- Consultar mis reseñas

- Consultar tutorias confirmadas
- Consultar tutorias Pendientes
- Consultar mis reseñas

- Editar Tutoria
- Editar Titulo
- Editar descripción
- Cambiar foto
- Editar Ubicación
- Editar costo de la tutoria
- Guardar y Publicar
- Editar Datos de contacto
- Eliminar Tutoria

- Consultar tutorias
- Consultar Tutor
- Consultar Certificaciones
- Consultar Reseñas